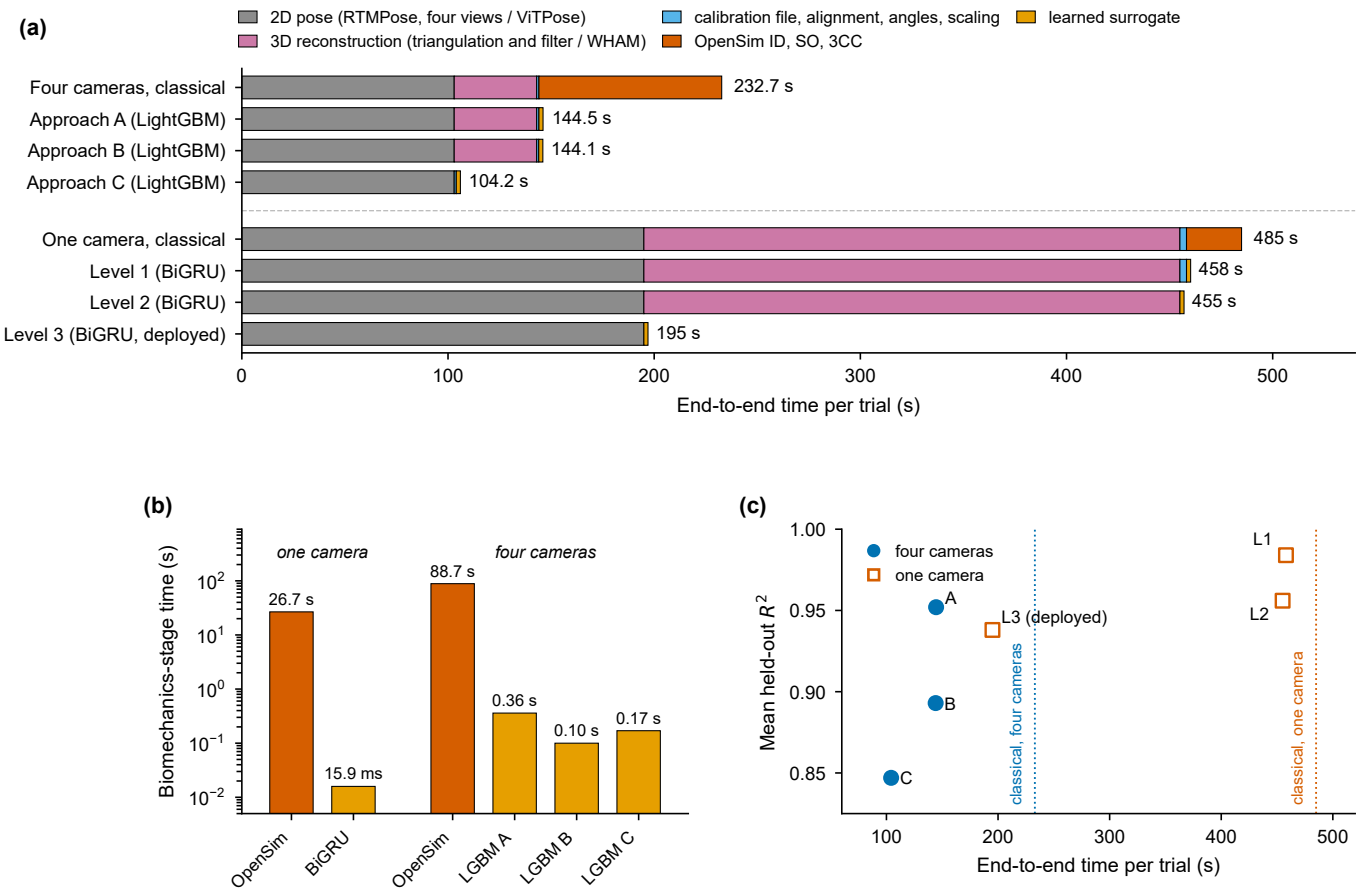

(a)
2D pose (RTMPose, four views / ViTPose)
3D reconstruction (triangulation and filter / WHAM)
calibration file, alignment, angles, scaling
OpenSim ID, SO, 3CC
learned surrogate
Four cameras, classical
Approach A (LightGBM)
Approach B (LightGBM)
Approach C (LightGBM)
One camera, classical
Level 1 (BiGRU)
Level 2 (BiGRU)
Level 3 (BiGRU, deployed)
232.7 s
144.5 s
144.1 s
104.2 s
485 s
458 s
455 s
195 s
0
100
200
300
400
500
End-to-end time per trial (s)
(b)
Biomechanics-stage time (s)
one camera
four cameras
26.7 s
15.9 ms
88.7 s
0.36 s
0.10 s
0.17 s
OpenSim
BiGRU
OpenSim
LGBM A
LGBM B
LGBM C
(c)
Mean held-out R^2
four cameras
one camera
A
B
C
L1
L2
L3 (deployed)
classical, four cameras
classical, one camera
1.00
0.95
0.90
0.85
100
200
300
400
500
End-to-end time per trial (s)